Geometry-MMALS G1 v1.1.3

Functional Memory Transport under Matched Remembered-Source Access

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Corrected C0 specification – build `functional-memory-transport-c0-r2`

Abstract

Geometry-MMALS G1 v1.1.1 isolated the context-to-route bridge under one frozen host ecosystem. Version 1.1.2 repaired sharp prototype-routing tails but did not outperform the low-capacity linear router on the preregistered held-out functional contrast. Version 1.1.3 therefore stops adding router capacity. It fixes one linear router and asks which memory object should constrain it during sequential learning. Treatments receive the same remembered identities, the same access to old images, the same memory batches, and matched compute, but deliberately preserve memory objects of different richness. The protocol compares current-only learning, replay, nominal route anchoring, functional transport, and joint functional memory. It also measures paired and distributional drift, old-angle forgetting, mediated-latent and output drift, prediction identity, accumulated regularized transport cost, and a covariance-aware compiled-memory diagnostic. Gradient-scale calibration is mandatory before the five-seed pilot. This is the final planned experiment in the fully frozen-host regime.

1 Scientific role and stopping rule

The v1.1.2 result indicates that context structure and router smoothness are not sufficient to produce a qualified functional bridge in the frozen ecosystem. v1.1.3 is therefore a closure experiment, not another attempt to increase router complexity. It asks whether a better-defined memory object preserves route-function behavior under the same frozen ecosystem.

After v1.1.3 the program leaves the fully frozen-host regime regardless of outcome. A positive result supports a narrow memory claim; a negative result records a limit of the frozen regime. No additional router, distance, or frozen-bank memory variant is planned.

2 Frozen ecosystem and reconstruction acceptance

All treatments reconstruct and then freeze

$$z_0 = P(x), \qquad c = C_\phi(z_0), \qquad u_h = g_h(z_0),$$

$$z_M(r) = \sum_{h=1}^{H} r_h u_h, \qquad \hat{y} = Q(z_M).$$

The sensory encoder, context encoder, host bank, synthesis normalization, classifier, and common cost matrix are shared. Only a linear router is trained:

$$r_\theta(c) = \mathrm{softmax}(Wc + b).$$

Every treatment starts from the same fresh initialization for a given seed, with router seed equal to model seed plus 1300. No trained v1.1.2 router is loaded.

Exact parameter hashes are exported for audit but are not portability gates. Reconstruction acceptance is metric-based: sensory test accuracy must be at least 0.92, compared with the v1.1.2 reference value 0.942. After treatment, every frozen-component delta must be at most 10^{-6}.

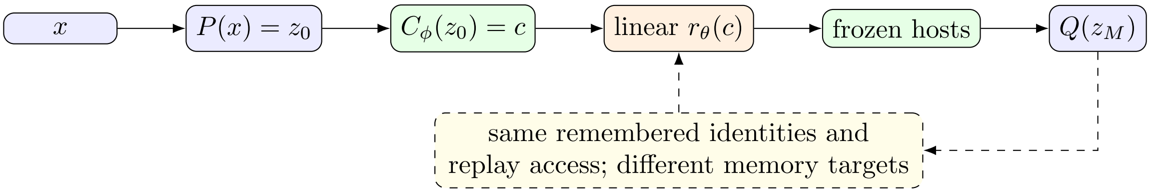

Figure 1: Only the memory objective changes. The entire functional ecosystem and router architecture are fixed.

3 Matched remembered-source access, not matched bytes

At each stage every treatment receives the current training subset and exactly 64 remembered source identities per previously learned angle. The identity list, old-image accesses, memory batches, current and memory forward counts, and optimizer-step counts are matched. No remaining historical examples are available.

This is a *matched remembered-source and replay-access budget.* It is not a byte-matched budget. M1 uses labels, M2/M3 preserve route targets, and M4 additionally preserves mediated-latent and output targets. The primary question deliberately compares richer memory objects for the same remembered identities. Estimated auxiliary target bytes are exported for transparency. A byte-matched sensitivity is optional, disabled, and secondary.

4 Acquisition snapshot contract

Let a be an angle, $\tau(a)$ its acquisition stage, s a later stage, and T the final stage. The original memory state is captured in this exact order:

1. complete the last optimizer step of stage $\tau(a)$;
2. deep-copy the router state immediately;
3. instantiate a frozen teacher from that copy;
4. compute teacher targets on the remembered identities of a;
5. evaluate the end of stage $\tau(a)$;
6. only then begin any operation belonging to stage $\tau(a) + 1$.

The notebook exports separate router-snapshot and teacher-target artifacts. Targets are never recomputed from a later model.

5 Treatments

Treatment	Current CE	Replay CE	Route	Functional OT	Latent	Output KL
M0 current only	yes	zero weight	no	no	no	no
M1 replay CE	yes	yes	no	no	no	no
M2 route anchor	yes	yes	yes	no	no	no
M3 functional transport	yes	yes	no	yes	no	no
M4 joint functional memory	yes	yes	no	yes	yes	yes

M0 still performs the matched old-memory forward pass but receives no old-memory gradient.

6 Exact losses and functional cost

Nominal anchoring acts in root-simplex coordinates:

$$\mathcal{L}_R = \frac{1}{2B} \sum_n \left\| \sqrt{r_s^{(n)}} - \sqrt{r_{\mathrm{ref}}^{(n)}} \right\|_2^2 .$$

Let d_z be the host-output dimension. The unnormalized host cost is

$$\widetilde{C}_{hk} = \frac{1}{d_z} \mathbb{E}_x \left[\|g_h(P(x)) - g_k(P(x))\|_2^2 \right] ,$$

and the shared functional matrix is

$$C^\star_{hk} = \frac{\widetilde{C}_{hk}}{\mathrm{median}_{i<j}\, \widetilde{C}_{ij}}, \qquad C^\star_{hh} = 0.$$

The paired functional loss is

$$\mathcal{L}_F = \frac{1}{B} \sum_n \mathrm{OT}_{C^\star,\epsilon_r} \left(r_s^{(n)}, r_{\mathrm{ref}}^{(n)} \right) ,$$

with $\epsilon_r = 0.05$ and 100 Sinkhorn iterations. Latent and output preservation are

$$\mathcal{L}_Z = \frac{1}{B} \sum_n \|z_{M,s}^{(n)} - z_{M,\mathrm{ref}}^{(n)}\|_2^2,$$

$$\mathcal{L}_Y = T_d^2 \, \mathrm{KL}(\mathrm{softmax}(\ell_{\mathrm{ref}}/T_d) \parallel \mathrm{softmax}(\ell_s/T_d)) ,$$

where $T_d = 2.0$ and the route temperature is 1.0.

7 Mandatory pre-pilot gradient calibration

Equal coefficients do not imply equal router-gradient pressure. One development calibration is mandatory before the pilot. It uses seed 0, eight preregistered probe batches, the same old-memory examples, and the same frozen ecosystem. After limited current-task drift, it measures

$$g_R = \mathrm{median}\, \|\nabla_\theta \mathcal{L}_R\|_2, \qquad g_F = \mathrm{median}\, \|\nabla_\theta \mathcal{L}_F\|_2.$$

The route coefficient is fixed at $\lambda_R = 0.5$. The proposed functional coefficient is

$$\lambda_F = \mathrm{clip}\left(\lambda_R \frac{g_R}{g_F}, 0.05, 2.0 \right) .$$

The calibration report, selected value, and report SHA-256 must be committed to the YAML before pilot execution. Pilot and qualification profiles refuse to start until calibration is marked `locked`. Coefficients are never adapted during the pilot.

8 Distributional transport and path diagnostics

For angle a after stage s,

$$\mu_{a,s} = \frac{1}{N} \sum_n \delta_{r_s(x_n,a)}.$$

The inner route-cloud ground cost is

$$D_{ij}^{(r)} = \mathrm{OT}_{C^\star,\epsilon_r}(r_i, r_j),$$

and the outer empirical transport is

$$\mathcal{T}_a^{s-1\to s} = \mathrm{OT}_{D^{(r)},\epsilon_c}(\mu_{a,s-1}, \mu_{a,s}),$$

with $\epsilon_c = 0.05$ and 100 iterations.

The accumulated regularized path cost and direct endpoint cost are

$$L_a = \sum_{s=\tau(a)+1}^{T} \delta_F(\mu_{a,s-1}, \mu_{a,s}), \qquad D_a = \delta_F(\mu_{a,\tau(a)}, \mu_{a,T}),$$

and the reported ratio is

$$E_a = \frac{L_a}{D_a + \varepsilon}.$$

E_a is a path-to-endpoint *regularized cost ratio*, not a geodesic excess. Raw entropic OT is not guaranteed to be a metric; therefore $E_a \geq 1$ is not guaranteed and values below one are possible.

9 Compiled-memory diagnostic

For root routes $y = \sqrt{r}$, the covariance-aware summary is $q_a^{\mathrm{cov}} = \mathcal{N}(m_a, \Sigma_a)$. The single baseline is the trace-matched isotropic Gaussian

$$q_a^{\mathrm{iso}} = \mathcal{N}(m_a, \sigma_a^2 I), \qquad \sigma_a^2 = \frac{1}{d_r}\,\mathrm{tr}(\Sigma_a).$$

The diagnostic reports

$$\Delta\mathrm{NLL} = \mathrm{NLL}(q^{\mathrm{cov}}) - \mathrm{NLL}(q^{\mathrm{iso}}),$$

where negative values favor covariance-aware compilation. This does not replace replay.

10 Preregistered constants

Element	Value
Remembered identities	64 per previous angle
Identity selection	fixed permutation, first 64 memory indices
Router initialization	model seed +1300
Route / distillation temperatures	1.0 / 2.0
Route Sinkhorn	$\epsilon = 0.05$, 100 iterations
Cloud Sinkhorn	$\epsilon = 0.05$, 100 iterations
New-angle margin	0.01
Sensory acceptance floor	0.92
Calibration seed / batches	0 / 8
λ_R	0.5
λ_F	locked after development calibration

11 Contrasts and decision rules

ID	Contrast	Metric	Favorable
A1	M3–M2	endpoint functional drift	< 0
A2	M3–M2	old-angle forgetting	< 0 (secondary)
B1	M4–M1	old-angle forgetting	< 0
B2	M4–M1	path cost ratio	< 0 (secondary)
S1	M4–M1	new-angle accuracy	$\lvert\Delta\rvert \leq 0.01$
D1	covariance–isotropic	held-out root NLL	< 0 (diagnostic)

A1 and B1 each require a five-seed 95% confidence interval with upper bound below zero and at least four favorable seeds. A2 is always reported but cannot substitute for A1.

Protocol-integrity gates cover frozen components, reconstruction acceptance, matched compute, exact remembered-source count, source-disjoint evaluation, acquisition ordering, and a locked calibration report. Failure blocks the associated claim.

12 Interpretation discipline

A positive A1 result supports only that functional transport preserves old route-function behavior better than nominal anchoring under matched remembered-source access and compute in a frozen ecosystem. A positive B1 result supports only that richer joint memory reduces forgetting beyond replay for the same remembered identities and old-image access.

The experiment does not claim byte efficiency, mature host specialization, positive backward transfer, operational superiority, transport metricity or optimality, geodesic motion, a learned Riemannian manifold, RL control, or quantum advantage.